

MEETING MINUTES

Project:	ELEN4000A/4011A Design II — Topic 2: Design and Simulation of a Nano-Grid for African Electrification
Supervisor:	Dr Mohammed Raees Dangor
Date:	Monday, 27 July 2026
Type:	Second supervisor meeting, Week 2
Attendees:	Dr Mohammed Dangor (Supervisor), Claude-Mary Moundzika-Kibamba, Boaz Reddy, Bruce Nkabinde, Dirren Moodley, Saien Chetty, Samuel Nthoroane, Sergio Lephadi, Shekhar Krishanlall

1. Site Updates & Group Coordination

- Groups presented site choices and geographic verifications across Sub-Saharan Africa.
- Dr Dangor noted that teams targeting the same country (e.g., Malawi) should check in with each other to ensure sites do not overlap, while encouraging teams to share relevant regional research.
- **Guidance on AI for Image Analysis:** When using AI tools to process satellite imagery or count roof structures, students must understand the underlying methodology and prompt logic, and manually double-check all counts for accuracy.

2. Structure Classification & Census Terminology

- Dr Dangor highlighted that terms like "household" describe social or demographic units rather than physical buildings. An external examiner reading "household" will assume formal housing.
- **Direction:** Use precise categories from official national census data (e.g., traditional mud-walled structures, thatched units, corrugated iron roofs) to accurately describe physical dwellings.

3. Scope Confirmation & Baseline Assumptions

- **Phased Electrification:** Projects do not need to electrify an entire settlement all at once. Sizing a design to cover a primary cluster first, backed by a clear engineering strategy for phased expansion, is fully acceptable.
- **Zero-Grid Baseline Rule:** All designs must assume zero existing electrification per the project brief. Any standalone solar panels or power infrastructure visible in satellite imagery must be explicitly documented as excluded from the design scope.

4. Eskom Case Study: Rural Electrification Lessons

- Dr Dangor shared a historical case study on Eskom's post-1994 rural electrification program to highlight key planning pitfalls:
 - **Capital Cost Overruns:** Connection costs per dwelling were initially budgeted at ~R3,000, but difficult terrain and line routing drove actual costs up to ~R15,000 (a 5x overrun).

- **Appliance & Consumption Gap:** Sizing and financial models assumed households would adopt standard AC appliances (TVs, fridges, stoves). In reality, residents owned very few appliances, leading to near-zero energy use and the collapse of the payback model.
- **Key Takeaway:** Load profiles and revenue models must reflect realistic rural appliance ownership and actual daily usage patterns rather than theoretical power supply.

5. Competitor Failure Analysis

- Reviews of existing off-grid energy solutions should focus explicitly on their documented failure modes and operational challenges (e.g., connection cost overruns, theft, maintenance barriers, or rigid central units), rather than only their successes.
- Demonstrating how a proposed design avoids these specific historical pitfalls provides strong technical justification.

6. Stakeholder Alignment & Local Leadership

- Field experience highlights the necessity of proper community engagement:
 - **Formal Leadership Engagement:** Always identify and account for recognized local administrative and traditional leaders. Skipping local leadership risks political pushback and community suspicion.
 - **Community Hubs:** Powering shared social spaces or local commercial nodes builds early community buy-in and demonstrates immediate system value.

7. Physical Security Design Considerations

- Off-grid solar installations face severe risks of theft and vandalism. Dr Dangor shared security insights from past field deployments:
 - Adding barbed wire to support poles often led to thieves cutting down the entire pole rather than climbing it.
 - Running security wire loops *inside* structural poles creates an effective tripwire that sets off an alarm if a pole is cut.
 - Enclosure boxes using magnetic reed switches reliably detect unauthorized opening.
- **Key Takeaway:** Most off-grid sites lack cellular or internet coverage. Security designs should rely on simple, hardwired analogue circuits triggering local alarms or lights rather than internet-dependent IoT devices.

8. Load Profile & ESMAP Tiering Methodology

- **Methodology:** Dr Dangor confirmed that two sizing approaches are equally valid: selecting a target ESMAP Tier (e.g., Tier 2) first and building an appliance list to match, or establishing an appliance profile first and mapping it to an ESMAP Tier. ESMAP literature directly links specific tiers to benchmark appliance profiles and daily usage windows.

9. Action Items Raised in the Meeting

- **Confirm Dwelling Terminology:** Reference official national census classifications to accurately categorize structure types in design documentation.

- **Evaluate Competitor Failure Modes:** Analyze existing off-grid commercial models and document where they struggled (e.g., high connection costs, theft, or maintenance issues) to demonstrate how the proposed design avoids these pitfalls.
- **Refine Load Profiles:** Define daily energy consumption, appliance power ratings (W), and time-of-use schedules aligned with target ESMAP access tiers—grounding load models in realistic rural appliance ownership rather than assuming standard AC usage.
- **Design Analogue Physical Security:** Outline simple, hardwired anti-theft measures (such as internal pole-cut sensing wire loops or magnetic box switches) that work reliably off-grid without requiring cellular or internet connections.
- **Map Local Leadership Structures:** Identify key local administrative and traditional leadership hierarchies to establish a clear strategy for community engagement and project buy-in.